

Data Preparation and Transformation

Scope. Three source files were profiled, corrected, enriched and joined into a single analytical dataset before any visual was built. This document records what was checked, what was changed and what was found, with screenshot evidence for each step. Two data defects were caught during verification and both changed how the dashboard was built.

1. Sources and storage

Three CSV files were loaded into SPICE, the in-memory store behind Amazon Quick, and a fourth dataset was created by joining them. All four are held in SPICE on a weekly refresh, which matches the stated requirement for weekly freshness and keeps the dashboard responsive without querying the source files on every interaction.

Dataset	Rows	Columns	Role
NovaTech_CRM_Deals	499	22	Closed sales opportunities across 85 accounts, the anchor table for the join
NovaTech_Marketing_Campaigns	2,240	23	Leads across six campaigns and five channels
NovaTech_Support_Tickets	3,000	25	Support tickets with priority, product area and sentiment
NovaTech_Unified	63,420	66	All three joined on the account identifier, used for account-level analysis

Data

DatasetsTopicsData sources

Create dataset

Search datasets by name

All datasets

Name		Owner	Last Modified	Action
 NovaTech_Support_Tickets	SPICE	Me	3 days ago	Chat
 NovaTech_CRM_Deals	SPICE	Me	3 days ago	Chat
 NovaTech_Unified	SPICE	Me	7 days ago	Chat
 NovaTech_Marketing_Campaigns	SPICE	Me	7 days ago	Chat

Evidence 1. The dataset library. All four datasets are held in SPICE rather than queried directly, which is what keeps the dashboard responsive under interaction. The unified dataset is built from the three sources rather than imported separately.

1.1 SPICE import verification

Each dataset was checked individually after import. The figures below come from the dataset summary page, which reports the ingestion result and the column count together. All three source files imported at 100% success with row counts matching the source exactly, and the joined dataset materialised at its expected 63,420 rows.

Dataset	Source file	Rows imported	Success	Columns	Row count against source
NovaTech_CRM_Deals	novatech_crm_deals.csv	499	100%	22	Exact match
NovaTech_Marketing_Campaigns	novatech_marketing_campaigns.csv	2,240	100%	23	Exact match
NovaTech_Support_Tickets	novatech_support_tickets.csv	3,000	100%	25	Exact match
NovaTech_Unified	Join of the three above	63,420	100%	66	Derived, see section 6

A note on the column counts. Row counts are the integrity check that matters on import: if a file loads short or long, something was lost, duplicated or misparsed, and all three match their source exactly. Column counts are deliberately not expected to match the raw files, because these are prepared datasets rather than raw copies. Each one carries its source columns plus the calculated fields added during preparation, listed in section 5, and minus any column dropped as unusable, such as the constant data_source field. Support tickets, for example, reports 25 columns because five calculated fields were

added on top of the retained source columns. The counts above are what SPICE actually holds and what the dashboard is built on.

Quick

Datasets

NovaTech_CRM_Deals

UdacityQuicksightL

NovaTech_CRM_Deals

Generate analysis

Chat

Edit dataset

NovaTech CRM deals — sales pipeline of 499 closed opportunities (315 Won, 184 Lost) across 85 accounts, created 2023-06-17 to 2025-01-25.

Summary

Refresh

Permissions

Usage

Columns (22)

Search

Name	Type	Description
account_id	String	
company_name	String	
industry	String	
company_size_tier	String	
annual_revenue_usd	Decimal	

About

SPICE

Size: 278.4 KB

Refresh

Completed

499 rows imported (100% success)

Last refresh at

September 11, 2026 at 10:12 AM GMT+1

Evidence 1a, NovaTech_CRM_Deals. 22 columns, SPICE storage, 499 rows imported at 100% success. The description line records the profile confirmed during verification: 499 closed opportunities, 315 won and 184 lost, across 85 accounts.

< Datasets

NovaTech_Marketing_Campaigns

UdacityQuicksightL

NovaTech_Marketing_Campaigns

Generate analysis

Chat

Edit dataset

Summary

Refresh

Permissions

Usage

Columns (23)

Search

Name	Type	Description
account_id	String	
lead_id	String	
campaign_name	String	
campaign_channel	String	
campaign_date	Datetime	
funnel_stage	String	
customer_segment	String	
days_since_last_engagement	Integer	
spend_novapulse	Integer	
spend_novaeedge	Integer	

About

SPICE

Size: 941.8 KB

Refresh

Completed

2240 rows imported (100% success)

Last refresh at

September 7, 2026 at 11:46 AM GMT+1

Access

Row-level security

Set up

Column-level security

Set up

Sources

novatech_marketing_campaigns.csv

Evidence 1b, NovaTech_Marketing_Campaigns. 23 columns, SPICE storage, 2,240 rows imported at 100% success, matching the 2,240 leads reported on the Marketing Funnel view.

< Datasets

NovaTech_Support_Tickets

UdacityQuicksightL

NovaTech_Support_Tickets

Generate analysis

Chat

Edit dataset

Summary

Refresh

Permissions

Usage

Columns (25)

Search

Name	Type	Description
account_id	String	
ticket_id	String	
ticket_created_date	Datetime	
ticket_resolved_date	Datetime	
priority	String	
product_area	String	
contact_channel	String	
reported_by	String	

About

SPICE

Size: 1.5 MB

Refresh

Completed

3000 rows imported (100% success)

Last refresh at

September 11, 2026 at 10:14 AM GMT+1

Access

Row-level security

Set up

Column-level security

Set up

Evidence 1c, NovaTech_Support_Tickets. 25 columns, SPICE storage, 3,000 rows imported at 100% success. The source file is named in the Sources panel, and resolution_hours is visible in the column list as one of the calculated fields added during preparation.

NovaTech_Unified

Generate analysis

Chat

Edit dataset

SummaryRefreshPermissionsUsage

Columns (66)

Search

Q

Name	Type	Description
company_name	String	
industry	String	
company_size_tier	String	
annual_revenue_usd	Decimal	
employee_count	Integer	
headquarters	String	
opportunity_id	String	

About

SPICE

Size: 85.3 MB

Refresh

Completed

63420 rows imported (100% success)

Last refresh atSeptember 7, 2026 at 5:16 PM GMT+1

Access

Row-level security

Set up

Column-level security

Set up

Evidence 1d, NovaTech_Unified. 66 columns drawn from all three sources, 63,420 rows, materialised in SPICE at 85.3 MB. This is the fan-out described in section 6, and it is the reason every figure on the dashboard is built with distinct counts rather than plain sums.

2. Data quality verification

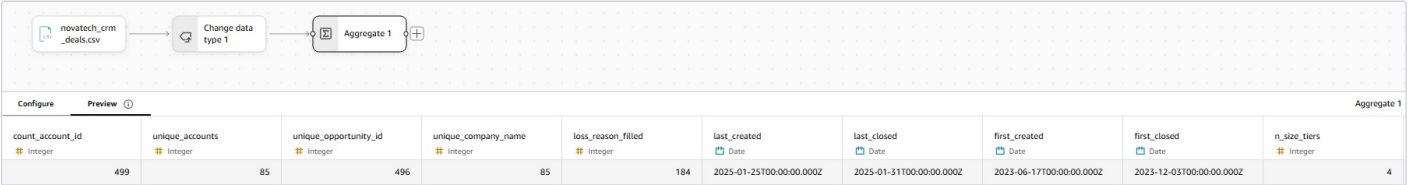
Every source was checked against the supplied data dictionary before transformation, using natural-language queries against the indexed data. Eight checks were run covering all three sources, targeting facts that can be proved: row counts, distinct values, date ranges and known null counts.

#	Knowledge base	Question asked	Expected from dictionary	Actual answer	Result	Notes
1	CRM Deals	How many records are in the CRM deals data?	499 rows	499 records returned	Pass	Also served as the load check, the file imported completely.
2	CRM Deals	How many deals were won and how many were lost?	Won 315, Lost 184, total 499	Same split returned with the breakdown	Pass	Confirms deal_stage has exactly two values and no nulls.
3	Marketing Campaigns	How many rows have a missing annual_income?	24 rows, 1.1%	24 rows with a null annual_income	Pass	Known gap, documented in the dictionary. Left as null rather than imputed.
4	Marketing Campaigns	How many leads have campaign_response = 1, and what is the response rate?	609 of 2,240 leads, 27.2%	Total 2,240, responders 609, rate 27.2%	Pass	Response rate reproduced independently before it was built as a KPI.
5	Support Tickets	How many support tickets have no ticket_resolved_date?	59 tickets, 2.0%	59 tickets with no resolved date	Pass	Cross-checked against the raw file: 3,000 minus 2,941 resolved equals 59.
6	Support Tickets	How many tickets are there for each priority level?	low 1,500, medium 1,050, high 400, critical 50	Identical distribution, total 3,000	Pass	Confirms the priority field is complete and correctly valued.
7	Reference documents	How many sales reps does NovaTech have, and how are they organised by region?	30 reps across East, Central and West	30 unique reps, 10 per region, no rep in two regions	Pass	Answered from the CRM data rather than the brief. Counts agree with the company background.
8	Reference documents	Who are the sales managers, and who do they report to?	Six managers reporting to the VP of Revenue	Named all six with their regions and five reps each. Stated that the reporting line is not in the dataset	Partial	The reporting line exists only in the company background document, which is not indexed in this workspace. Q said so rather than guessing, which is the correct behaviour.

3. Structural profiling, and two defects found

Beyond the question-and-answer checks, an aggregate probe was run inside the dataset editor to verify keys, formats and ranges in a single pass.

Check	Result	Dictionary states	Verdict
Rows loaded	499	499 rows	Match
Distinct accounts	85	85 accounts	Match
Distinct company names	85	One per account	Match
Distinct opportunity IDs	496	Expected 499	Mismatch , 3 IDs reused across different accounts
Rows with a loss reason	184	Blank for 315 Won deals	Match, 499 minus 184 equals 315
Account IDs matching ACCT- <i>nnn</i>	499	Format ACCT-XXX	Match, 100% well formed
Opportunity IDs matching OPP- <i>nnnnn</i>	499	Format OPP-XXXXX	Match, 100% well formed
Deal creation date range	2023-06-17 to 2025-01-25	Same range	Match
Distinct company size tiers	4	Enterprise, Large, Medium, Small	Noted , the support system records only 3 sizes
Distinct industries / HQ countries	10 / 15	10 / 15	Match, includes the 'Philippines' spelling present in the source



Evidence 2. Aggregate probe output. 499 rows, 85 accounts, 496 distinct opportunity identifiers against 499 deals, 184 rows carrying a loss reason, and the full creation and close date ranges.

Defect 1, opportunity identifiers are not unique

The dictionary implies one identifier per deal, which would give 499. The data returns 496, so three identifiers are reused on different accounts. A deal is therefore only safely identified by account and opportunity together. This matters because any count built on the opportunity identifier alone would silently under-report by three.

Defect 2, the two systems disagree on customer size

The CRM records four company size tiers while the support system records three. The same customer can therefore carry a different size label depending on which system is asked. Segment filtering on the dashboard is built to respect this rather than assume the two are interchangeable.

4. Data type corrections

Every column arrived as text. Dates, currency and counts were retyped so that date arithmetic, sorting and aggregation behave correctly. One correction was made twice: deal value was auto-detected as an integer, which would truncate cents on a future refresh, and was reset to decimal.

Column	Change	Reason
deal_created_date, deal_closed_date, ticket_created_date, ticket_resolved_date	Text to Date, format yyyy-MM-dd	Required for date arithmetic and for time axes to sort chronologically
deal_value, annual_revenue_usd, error_rate_pct	Text to Decimal	Currency and rates carry decimals. Deal value had been auto-detected as an integer, which would truncate on refresh
employee_count, list_price, users_affected, downtime_minutes, payment_impact, security_incident, data_loss	Text to Integer	Whole numbers and binary flags, needed for aggregation rather than as labels
data_source	Removed	Constant value with no analytical use

novatech_sup
port_tickets....

Change data
type 1

Aggregate 1

Configure Preview

Select columns to change data type

Search columns

- ☐ account_id
- ☐ ticket_id
- ☒ ticket_created_date
- ☒ ticket_resolved_date
- ☐ priority
- ☐ product_area
- ☐ contact_channel
- ☐ reported_by
- ☐ customer_tier
- ☐ company_size
- ☐ region
- ☒ users_affected
- ☒ error_rate_pct
- ☒ downtime_minutes
- ☒ payment_impact
- ☒ security_incident
- ☒ data_loss
- ☐ customer_sentiment
- ☐ tickets_last_30_days
- ☐ data_source

9 columns with data type changed

COLUMN NAME	ORIGINAL DATA TYPE	NEW DATA TYPE	
ticket_created_date	String	Date	☒ Date format is set to yyyy-MM-dd.
ticket_resolved_date	String	Date	☒ Date format is set to yyyy-MM-dd HH:mm:ss.
users_affected	String	Integer	☒
error_rate_pct	String	Decimal	☒
downtime_minutes	String	Integer	☒
payment_impact	String	Integer	☒
security_incident	String	Integer	☒
data_loss	String	Integer	☒

Evidence 3. Change data type step on the support tickets dataset, nine columns retyped from text to Date, Integer and Decimal, with the date format set explicitly rather than inferred.

5. Calculated fields

Eleven fields were added carrying business logic. Two of them exist because of data defects rather than analytical need: sentiment_clean, because blank sentiment values were stored as empty strings rather than nulls and would have been dropped silently from the sentiment breakdown, and the pair of format checks used during verification and removed before publishing.

Field	Dataset	What it computes	Why it exists
days_to_close	CRM Deals	Whole days between deal creation and close	Sales cycle length, requested in the brief and used on the Days to close trend chart
is_won	CRM Deals	1 for a won deal, 0 for a lost deal	Its average is the win rate, and it is the filter behind every won-only revenue figure
campaign_roi	Marketing Campaigns	Attributed revenue minus spend, divided by spend	Campaign return, the measure behind the ROI by campaign chart
is_closed_won	Marketing Campaigns	1 where the funnel stage is Closed Won	Enables lead-to-deal conversion to be measured from the marketing side
funnel_order	Marketing Campaigns	Ranks the funnel stages 1 to 5	Forces the funnel to display in business order instead of alphabetically
total_product_spend	Marketing Campaigns	Sum of the three product spend columns	Single spend figure per lead
resolution_hours	Support Tickets	Hours between ticket creation and resolution	The measure behind every resolution time figure on Customer Health
is_unresolved	Support Tickets	1 where no resolved date exists	Drives the Unresolved KPI card
sentiment_clean	Support Tickets	Replaces null or blank sentiment with 'not recorded'	Blanks were stored as empty strings, not nulls, so they would have been silently dropped from the sentiment donut
is_negative	Support Tickets	1 where sentiment is negative	Its average is the negative sentiment rate used in the at-risk logic
is_high_impact	Support Tickets	1 where the ticket involves payment impact, a security incident or data loss	Separates business-critical tickets from routine volume

novatech_sup
port_tickets...

Change data
type 1

Add
calculated...

Select
columns 1

ConfigurePreview

5 calculated columns

+ Calculated column

Search columns

Column name	Calculation definition	Actions
resolution_hours	dateDiff([ticket_created_date], [ticket_resolved_date], "HH")	
is_unresolved	ifelse(isNull([ticket_resolved_date]), 1, 0)	
sentiment_clean	ifelse(isNull([customer_sentiment]) OR strLen(trim([customer_sentiment])) = 0, 'not recorded', [customer_sentiment])	
is_negative	ifelse([customer_sentiment] = 'negative', 1, 0)	
is_high_impact	ifelse([payment_impact] = 1 OR [security_incident] = 1 OR [data_loss] = 1, 1, 0)	

Evidence 4. Calculated columns on the support tickets dataset with their definitions. resolution_hours drives every resolution time figure on the dashboard, and sentiment_clean prevents blank sentiment values disappearing from the breakdown.

Amazon Quick

Datasets

NovaTech_Marketing_Campaigns

FileDataHelp

TRANSFORMOUTPUT

STEPS

Steps

Select a step to add a transformation

Search

INPUT

Add data

COLUMN TRANSFORMATIONS

Add calculated columns

Change data type

Rename columns

Select columns

COMBINE TRANSFORMATIONS

Append

Join

OTHER TRANSFORMATIONS

Aggregate

Filter

Pivot

Unpivot

novatech_mar
keting_camp...

Change data
type 1

Rename
columns 1

Add
calculated...

Select
columns 1

ConfigurePreview

4 calculated columns

+ Calculated column

Search columns

Column name	Calculation definition	Actions
campaign_roi	((revenue_attributed) - (campaign_spend)) / (campaign_spend)	
ix_closed_won	ifelse(funnel_stage = 'Closed Won', 1, 0)	
funnel_order	ifelse(funnel_stage = 'Prospect', 1, (funnel_stage = 'Lead', 2, (funnel_stage = 'Qualified Lead', 3, (funnel_stage = 'Opportunity', 4, 5)	
total_product_spend	(spend_novapulse) + (spend_novaeagle) + (spend_addons)	

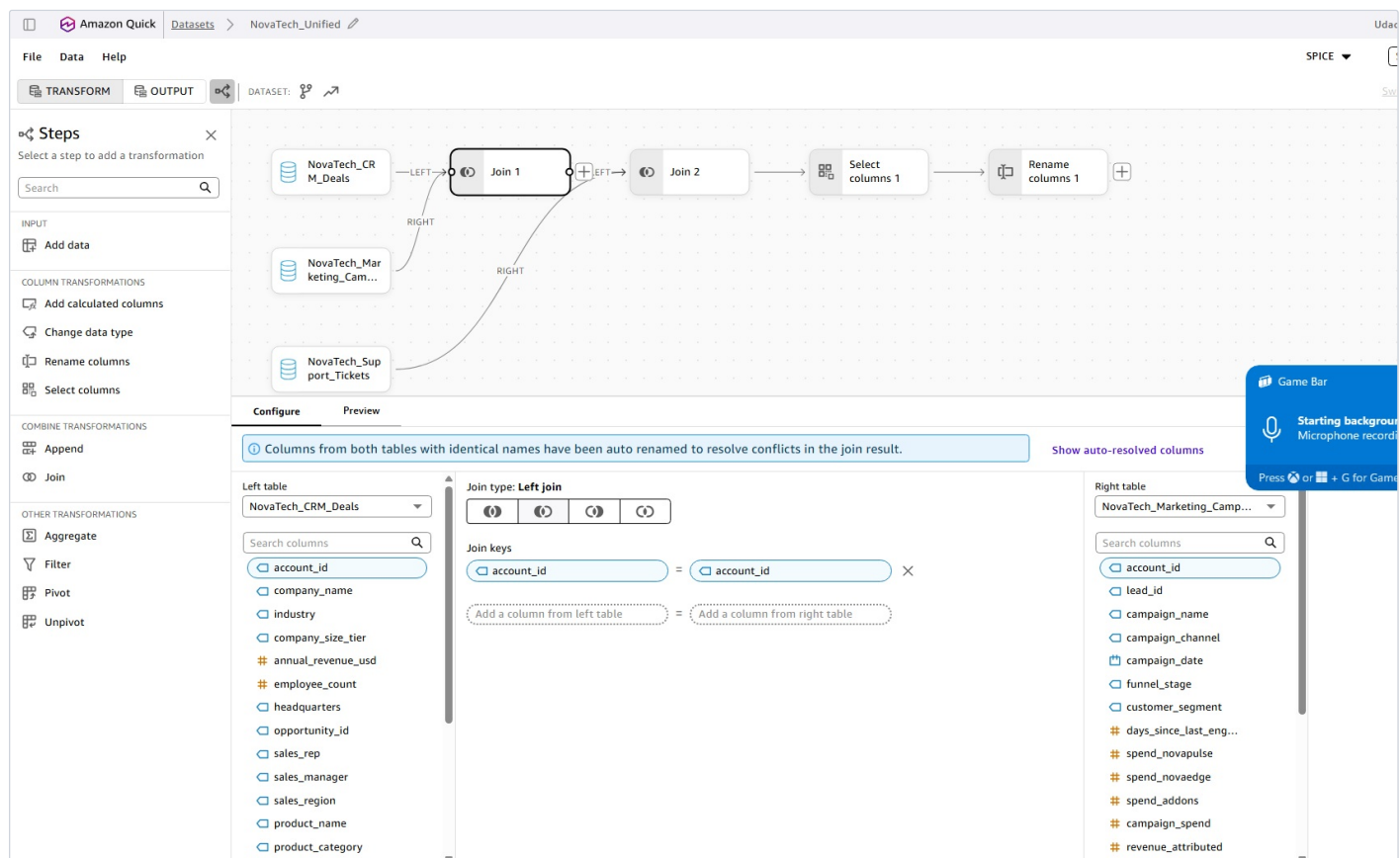
Evidence 5. Calculated columns on the marketing dataset. campaign_roi is the measure behind the ROI by campaign chart, and funnel_order forces the funnel stages to display in business sequence rather than alphabetically.

6. Join strategy

The three sources share one field, the account identifier, which appears in all of them. That field is what makes it possible to follow a single customer from the campaign that generated the lead, through the deal, to the support tickets that came afterwards.

Step	Left table	Join type	Right table	Key
Join 1	NovaTech_CRM_Deals (<i>anchor</i>)	Left join	NovaTech_Marketing_Campaigns	account_id = account_id
Join 2	Result of Join 1	Left join	NovaTech_Support_Tickets	account_id = account_id

Why the CRM is the anchor. Revenue is the spine of the analysis. Anchoring on deals keeps every deal in the result whether or not the account has marketing or support activity, which is what makes it possible to see that one account holds the highest revenue in the business while never having been marked as converted in marketing.



Evidence 6. Join diagram. CRM deals anchors the pipeline, marketing campaigns joins first, support tickets second, followed by column selection and renaming.

SPICE Save & publish Cancel

Switch back to legacy connections

Dataset: NovaTech CRM Deals

Join 1

Columns from both tables with identical names have been auto renamed to resolve conflicts in the join result.

Left table: NovaTech_CRM_Deals

Search columns: account_id, company_name, industry, company_size_tier, annual_revenue_usd, employee_count, headquarters, opportunity_id, sales_rep, sales_manager, sales_region, product_name, product_category, list_price, deal_stage, deal_created_date, deal_closed_date, deal_value, loss_reason, days_to_close, is_won

Join type: Left join

Join keys: account_id = account_id

Right table: NovaTech_Marketing_Campaigns

Search columns: account_id, lead_id, campaign_name, campaign_channel, campaign_date, funnel_stage, customer_segment, days_since_last_eng..., spend_novapulse, spend_novaeage, spend_addons, campaign_spend, revenue_attributed, web_visits_per_month, campaign_response, market_source_code, complaint_flag, campaign_roi, is_closed_won, funnel_order, annual_income, household_size, total_product_spend

Auto-resolved columns

Columns from left table	Columns from right table
ORIGINAL NAME	NEW NAME
account_id	account_id.1
account_id	account_id.2

Evidence 7. Join 1 configuration. Left table NovaTech_CRM_Deals, left join, keys account_id to account_id. The calculated fields days_to_close and is_won are visible at the foot of the left column list, carried through from the earlier transformation step.

The fan-out problem, and how it is handled

Joining one deal record to many support tickets multiplies rows. The three sources hold 499, 2,240 and 3,000 rows, and the joined result holds 63,420. Left unhandled, any total built on that dataset would be inflated by however many tickets and leads each account happens to carry.

Cleaning the fan-out out of the data is not the right fix, the row multiplication is a legitimate consequence of the join. It is handled at the analysis layer instead: every figure on the dashboard uses distinct counts and level-aware aggregation so each deal, lead and ticket is counted once. Totals were then checked back against the raw files. Total tickets reads 3,000, total deals 499, total leads 2,240, all matching the sources exactly.

NovaTech_Unified

Generate analysis

Chat

Edit dataset

SummaryRefreshPermissionsUsage

Columns (66)

Search

Name	Type	Description
company_name	String	
industry	String	
company_size_tier	String	
annual_revenue_usd	Decimal	
employee_count	Integer	
headquarters	String	
opportunity_id	String	
sales_rep	String	
sales_manager	String	
sales_region	String	
product_name	String	
product_category	String	
list_price	Integer	
deal_stage	String	
deal_created_date	Datetime	
deal_closed_date	Datetime	
deal_value	Decimal	
loss_reason	String	
days_to_close	Integer	

About

SPICE

Size: 85.3 MB

Refresh

Completed

63420 rows imported (100% success)

Last refresh atSeptember 7, 2026 at 5:16 PM GMT+1

Access

Row-level security

Column-level security

Set up

Set up

Sources

NovaTech_Support_Tickets

NovaTech_CRM_Deals

NovaTech_Marketing_Campaigns

Usage

Analyses (0)

Dashboards (0)

Datasets (0)

Evidence 8. The unified dataset in SPICE, 66 columns drawn from all three sources, with its three source datasets listed.

All figures verified against the source CSV files and the supplied data dictionary. Screenshots are unedited captures from the dataset editor.